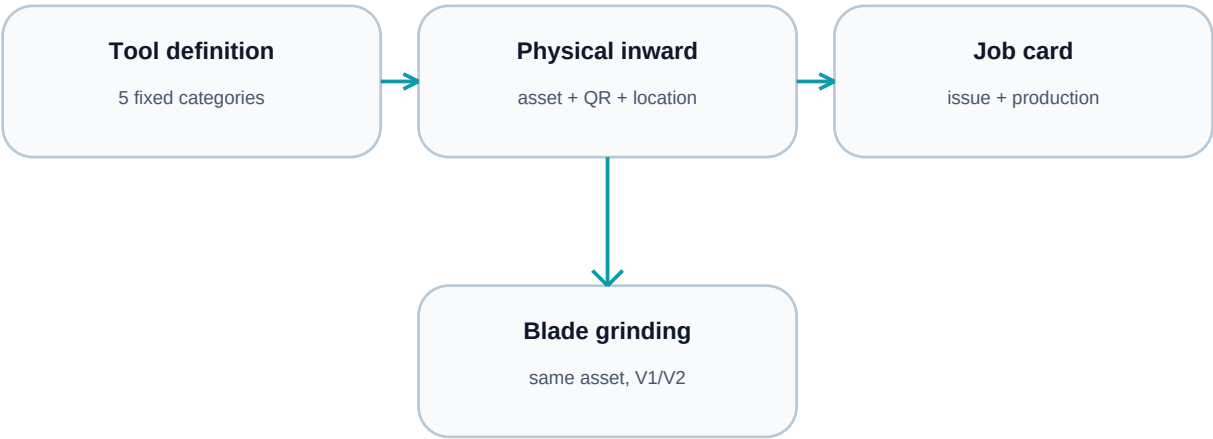


Tooling Lifecycle

Client operating report and user guide

From approved tooling master to physical QR asset, production issue, blade grinding, and trace reporting.



Release scope

This report explains the tooling and specification-sheet flows now available in the ERP. It is written for the client, store, production, quality, and supervisory teams. Technical implementation details are intentionally kept out of the operating instructions.

At a glance

Control	What the user gets
Five fixed categories	Notch, Blade, Holder, V + Flat, Punch
Physical identity	One asset number and QR value for every inwarded unit
Trace	Issue, return, maintenance, grinding, job-card output
Spec sheet	Approved values only, with mandrel-linked tube selection
Suggestions	Removed from the page and calculation flow

1. Tooling master

Open Masters > Tools. The five categories are fixed. The plant team can create multiple tool definitions under every category, but cannot create a sixth category.

Category	Fields maintained
Notch	Type, thickness, design, degree
Blade	Type, thickness, height, length
Holder	Thickness, height, length
V + Flat	Length, thickness
Punch	Punch option

Adding a definition

- Select Add Tool.
- Choose one of the five fixed categories.
- Enter the category points and status.
- Save. The definition becomes available to the specification sheet while active.

Editing approved dropdown values

Use the Editable dropdown registry to maintain tool Type, Design, Degree, Punch, and process Direction values. Values can be renamed, discontinued, and reactivated. Notch distance and depth are numeric inputs, not dropdown masters.

Important

Master definition and physical asset are separate records. Updating a definition does not erase the saved snapshot or history of a tool that was already inwarded.

2. Physical inward and QR identity

Use Inward beside the tool definition. Enter receipt date, quantity, and a Location Master position. The system expands the quantity into individual physical assets.

Generated information	Example
Asset number	TA-260714-001
QR value	hariom://tool/TA-260714-001
Starting status	Available
Grinding version	V0
Location	Tool Rack A

Example: two plain blades received on 14-Jul-2026 create two separate assets. They share the same definition but each has its own location, status, issue history, usage count, and production output.

3. Physical lifecycle

The ledger shows only the actions valid for the current physical status.

Status	Use
Available	Ready to issue, maintain, grind (blade), or scrap
Issued	Assigned to a job card and stage; return after use
Maintenance	Temporarily with maintenance; complete when usable
Grinding out	Blade sent for sharpening; grinding return completes the cycle
Scrap	Permanently removed from use; history retained

Issue to production

- Search or scan the QR/asset number.
- Select Issue.
- Choose a released job card and production stage.
- Save. The asset becomes Issued and the job card stores the exact asset ID.

Blade grinding

Return the blade if it is issued, select Grinding out, send it for sharpening, then select Grinding return. The same asset number remains in use and the version increments from V0 to V1, then V2, and so on.

A non-blade tool cannot use the blade grinding action.

4. Production output and trace

In the Winder or Process job-card section, select an available QR asset and issue it directly to the stage. At completion, actual output is recorded against open assignments even after a browser refresh. A retry cannot double-count the same completion.

Trace point	Example
Job card	JC-2026-014
Stage	Notching
Physical asset	TA-260714-001
Good output	1,200 tubes
Scrap output	15 tubes
Next lifecycle	Returned, then grinding V1

This answers how much each blade produced, how much scrap was associated with it, which job cards used it, and how many grinding cycles it completed.

5. Specification sheet

Open Specifications > New Specification. The sheet uses active tool definitions and active process option values. Discontinued records are not shown.

The notch section contains exactly eight process fields

- Notch type
- Notching blade
- Notching holder
- V + Flat
- Punch
- Notch direction
- Notch distance
- Notch depth

The first five are tool selections. Direction is an editable dropdown. Distance and depth are numeric inputs. Tool points are displayed from the master and are not retyped in the specification sheet.

Mandrel and tube rule

Select a mandrel first. The tube-size picker then becomes active and shows only tube IDs within plus or minus 1 mm of the selected mandrel. This keeps the sheet aligned with the physical mandrel choice.

Suggestions are removed

The specification sheet no longer shows suggestion cards or runs the removed suggestion calculation route. The user builds and saves the approved recipe directly from the masters and entered production values.

Browser evidence - searchable mandrel, filtered tube control, and no suggestion card (1 of 3)

NEW SPECIFICATION SHEET

Save Draft

Client requirement first. Everything else derives from it.

Pick customer, mandrel, tube, target weight, and CS first. Then build the paper mix and the manufacturing outputs will follow from one workbook rule.

GLUE 15.0% PARCHMENT 1.5% WET DIVISOR 0.910

DESIGN SPECIFICATIONS

CLIENT REQUIREMENT

Start with the commercial ask. The whole sheet derives from this block.

CUSTOMER MASTER TUBE SIZE MANDREL

TUBEOS
WORKSPACE
Specifications

DESIGN > SPECIFICATIONS Guide

MATERIAL RULE SHEET

One formula, one parchment gate, one adhesive split.

Wet target = dry target + 0.910. Glue and parchment are percentages of client dry

target. The glue and parchment are percentages of client dry target. The glue and parchment are percentages of client dry target.

BOOKS OPEN OPEN OWNER PLANT A Logout

Select customer MND-1... ALLOWED Yes

TUBE SIZE

110 x 122 x 149.9

TARGET DRY

WEIGHT

289.8 G

REQUIRED CS

KGF

Only tube IDs within +/- 1 mm of the selected mandrel are shown.

SHEET
REFERENCE**MND-
170905-
110.65-
110X122**

Pick
customer +
tube +
mandrel

MANDREL
ID BAND**110.65
mm**

Min 110.55
· Max
110.75

RECIPE-LED
OD**110.65
mm**

Wall 0.0000
mm

WET
DIVISOR**0.910**

Default
0.91 from
9%
moisture
loss

TARGET WET /
DRY**0.00 /
289.81
g**

Dry target
289.81 g with
divisor 0.910

TARGET
FORMULA
SPLIT**43.47
g glue ·
4.35 g
parchmen**

0.00 g
required
paper

PREDICTED
WET / DRY**0.00 /
0.00 g**

No recipe
applied yet

TUBE DRY
BAND**286.81
-
292.81
g**

Apply a
recipe to
compare
against the
min/max
band

ALLOWED PARCHMENT FAMILIES

Choose the family pool once. Vendor and color stay downstream.

AMMA

CHINA

SAGAR

0
SELECTED

ADHESIVE BREAKDOWN

Pick 2-3 masters. Ratios must total 100% of the fixed glue band.

RATIO

100%

Tl-4 (20100)

30

0.00

Remove

Vinsol (30100) v

70

0.00 g

Remove

Adhesive weight updates from the live paper total only.

Add Component

RECIPE MIX

Use three to five papers, stay under eighteen plies, and let the live mix drive the weights.

PAPERS

APPLIED COMBO RULE

Build the live recipe from approved paper, adhesive, and parchment masters.

3-5 PAPERS · 18 PLY MAX

Target wet = target dry ÷ 0.910. Glue and parchment are fixed from client dry weight; paper fills the remaining wet target.

PAPER TOTAL

0.00 g

CURRENT RECIPE WET / DRY

0.00 / 0.00 g

Target wet 0.00 g

DRY VS BAND

0.00 g

CODE	VARIETY	GSM	BF / PLY	THICK / PLY	WEIGHT	PLY	PLY NO.	ACTION
-	Select paper variety v	0	0.00 LOCKED	0.0000 mm Bulk 0.00 - Ply bond 0.00	0.00		1	Remove
TOTAL-ALL-PLY		0	0.00	0.0000	0.00	1	-	-

Paper rows drive wall thickness, tube paper weight, and the manufacturing output.

Add recipe row

LIVE RECIPE SUMMARY

Select customer

110 × 122 × 149.9

ONE BAMBOO YIELD

0 pcs

0 mm selected · 1520 mm usable

REQUIRED PAPER

0.00 g

0.00 g wet target

RECIPE STATUS

0.00 g

Select paper rows to build the live recipe against the target.

Recalculating recipe math after input settles. You can keep typing.

Hari Om Paper ERP | Tooling lifecycle guide

Page 6

MANUFACTURING MATRIX

ID comes from mandrel, OD comes from wall, and bamboo output follows the live recipe.

Bamboo weight is now the predicted recipe output for the usable cut length after trim loss.

MANDREL ID BAND

110.65 mm

Min 110.55 · Max 110.75

WALL / OD

0.0000 / 110.65 mm

OD = ID + 2 × wall

CURRENT RECIPE TUBE

0.00 / 0.00 g

Target wet 0.00 g · Paper 0.00 g · Glue 0.00 g

BAMBOO WET / DRY

0.00 / 0.00 g

0 pcs from 1520 mm usable

MANUFACTURING SPECIFICATION

Average is the live working size. Min/max stay tied to mandrel tolerance and recipe wall.

PARAMETER	MIN	AVG	MAX
I.D	110.55 mm	110.65 mm	110.75 mm
O.D	110.55 mm	110.65 mm	110.75 mm
Thickness	0.00 mm	0.00 mm	0.00 mm
Bamboo LT	0 mm	0 mm	10 mm
Tube Dry	287 g	290 g	293 g
Tube Wet	315 g	0 g	322 g

BAMBOO AND OUTPUT LOGIC

This block is what moves downstream into the job card.

SELECTED BAMBOO

0 mm

1520 mm usable after trim

TUBES / BAMBOO

0 pcs

150 mm tube length

PAPER / MM

0.0000 g

0.00 g required paper

NOTCH + TOOLING

Only the downstream tooling fields that still matter to the job card and print sheet.

EXPAND WHEN NEEDED

PACKING

Compact packing handoff only: box, plastic, fadda, and per-box counts.

EXPAND WHEN NEEDED

VALIDATION

Final footer and release checks only. Keep this out of the way until the sheet is ready.

EXPAND WHEN NEEDED

6. Reports and rejection trace

Open Reports > Tooling. Use the report for current physical status and output by asset.

- Total assets and counts by status.
- Category-level physical stock.
- Searchable physical ledger with location and QR value.
- One-click QR label printing and complete asset history.
- Lifetime output plus good/scrap output for each grinding version.
- Scrap quantity, grinding version, and current job assignment.
- Lifecycle history for inward, issue, return, maintenance, grinding, and scrap.

Customer rejection investigation

- Start with the order, finished-good lot, or job card.
- Read the physical tool asset IDs on the completed job-card stage.
- Open each asset from the tooling ledger.
- Review inward, location, issue/return, maintenance, grinding versions, and production output.
- Compare the asset output with the job-card and QC records.

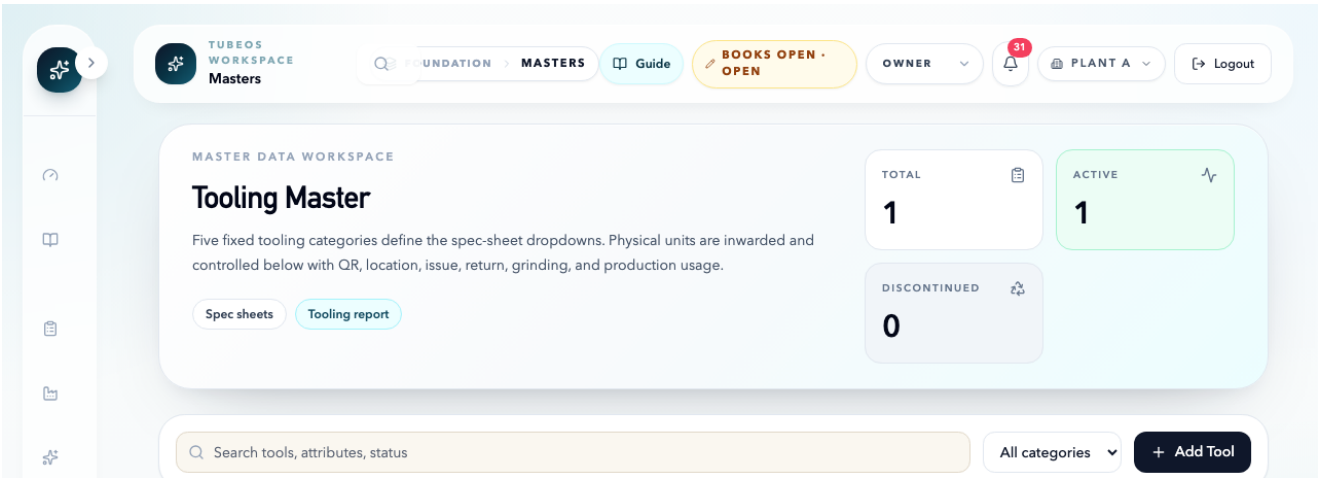
The result is a forward and backward trace from customer rejection to job card, physical tool, inward record, and maintenance/grinding history.

7. Daily checklist

- Maintain definitions and approved dropdown values before use.
- Inward every physical unit and assign its Location Master position.
- Apply the QR label.
- Issue and return the exact asset on every job card.
- Use grinding only for blades.
- Complete the job card so output is recorded.
- Review the tooling report at shift or month end.
- Discontinue or scrap records instead of deleting history.

Browser evidence - tooling master, option registry, and QR ledger

Tooling master and physical ledger (1 of 5)



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TOOL	CATEGORY	STATUS	USAGE	ACTIONS
Double Punch 5X10mm[center dist. 30mm] Seeded from Sanathan Polycoat P workbook	Punch	ACTIVE	0 spec/job logs	Edit Inward Discontinue Disable

EDITABLE DROPDOWN REGISTRY

Tool attributes
Only these option values feed the five tooling definitions and the notch process fields.

Notch

type

New option

+ Add

PHYSICAL TOOL CONTROL

QR asset ledger

TOTAL
0

AVAILABLE
0

ISSUED
0

GRINDING
0

Tooling master and physical ledger (3 of 5)

BLADE · TYPE

Plain

ACTIVE

Edit

Discontinue

BLADE · TYPE

Full Serration

ACTIVE

Edit

Discontinue

NOTCH · DEGREE

55

ACTIVE

Edit

Discontinue

NOTCH · DESIGN

Plain

ACTIVE

Edit

Discontinue

BLADE · TYPE

Half Serration

ACTIVE

Edit

Discontinue

NOTCH · DEGREE

50

ACTIVE

Edit

Discontinue

NOTCH · DEGREE

60

ACTIVE

Edit

Discontinue

NOTCH · DESIGN

Step

ACTIVE

Edit

Discontinue

Inward a physical unit against a definition, assign its Location Master position, and use the QR asset number for issue, return, grinding, and trace reports.

	NOTCH · NOTCH_DIRECTION Clockwise ACTIVEEditDiscontinue	NOTCH · NOTCH_DIRECTION Anticlockwise ACTIVEEditDiscontinue	
	NOTCH · TYPE Bottom LHS ACTIVEEditDiscontinue	NOTCH · TYPE Bottom RHS ACTIVEEditDiscontinue	
	NOTCH · TYPE Top RHS ACTIVEEditDiscontinue	PUNCH · PUNCH Single ACTIVEEditDiscontinue	
	PUNCH · PUNCH Double ACTIVEEditDiscontinue	PUNCH · PUNCH N/A ACTIVEEditDiscontinue	

Save

Edit

Discontinue

Save

Edit

Discontinue

PHYSICAL REGISTER

Inwarded tools and lifecycle

Q Search QR / asset no.

Scan QR

Open tooling report

ASSET / QR	DEFINITION	STATUS	LOCATION	GRINDING	PRODUCED	ACTION
------------	------------	--------	----------	----------	----------	--------

LEDGER

Recent Tool Events

Open report

TIME	TOOL	EVENT	SOURCE	REFERENCE
------	------	-------	--------	-----------

Browser evidence - job-card physical tool trace

Supervisor job card and physical tool snapshot (1 of 5)



TUBEOS
WORKSPACE
Dashboard

[OVERVIEW](#)
[DASHBOARD](#)
[Guide](#)

BOOKS OPEN · OPEN

OWNER ▾


31


PLANT A ▾

[Logout](#)

Supervisor Paper-to-ERP Entry

Supervisors load a planned card, confirm the auto-filled setup, and enter only actual run data from the floor sheet.

Load Job Card


Load

JC-224D8353 · Cutover Customer 170905 04	PLANNED
JC-B7B3D4A0 · Cutover Customer 170905 04	PLANNED
JC-E05B52DA · Cutover Customer 170905 04	PLANNED
JC-B1D904E6 · Cutover Customer 170905 04	PLANNED
JC-43575B3A · Cutover Customer 170905 04	PLANNED
JC-5613FD2A · Cutover Customer 170905 03	COMPLETED

Fast Entry Rules

Machine, shift, spec, mandrel, customer, order, parchment, and packaging all come from planning plus the approved setup snapshot.

Supervisors should only fill actual times, counts, measurements, reject reasons, and signatures.

A released job becomes floor-executable only after planner schedules its active stage into the next three days with a machine and shift.

SA

PRODUCTION EXECUTION

JC-224D8353

Shop-floor view driven from the saved manufacturing matrix. Assigned winder, bamboo target, recipe path, and current stage logging stay on one card so the operator and supervisor are reading the same truth.

CUSTOMER	PRODUCT	SPEC REF	SIZE
Cutover Customer 170905 04	FG-125-140-94	-	125 x 140 x 93.75
CURRENT STAGE	RELEASE QTY	ASSIGNED WINDER	PCS / BAMBOO
WINDER	64 pcs	Cutover Winder Low A	16
BAMBOO LENGTH	PARCHMENT	RELEASE LOT	SALES LINE
1540 mm	-	072a42b7	6ff9e64b

TARGET BAMBOO

4

OPEN SEGMENTS

1

CURRENT MACHINE

Cutover Winder
Low A

SHIFT SLOT

-- · 2026-05-25

PLANNER GATE

Blocked until planner slot is valid

Schedule machine, shift, and a planner date in the next 3 days before floor entry.

EXECUTION TARGET

4 bamboo • 16 pcs/bamboo • 1540 mm

Tube 44.32 / 48.7 g • Bamboo 709.07 / 779.19 g

QUEUED

SCAN ENTRY

Phone scan opens the live stage-entry screen for this exact job card on the running ERP host.

/production/entry/224d8353-48d8-4#6-915d-bfd0110a2024

CURRENT STAGE LOGGING

WINDER actual output capture

Enter only the live output, rejects, timings, and measured dimensions for the active segment. Manufacturing recipe, bamboo math, and packaging truth stay readonly on the right.

Open segment

WINDER SECTION

QUEUED

PLANNED WINDER Unscheduled	PLANNED SHIFT -	OPERATOR NAME -
SUPERVISOR SIGN -	QC SIGN -	START TIME -
END TIME	CYCLE TIME	PHYSICAL TOOLS

MANUFACTURING TRUTH

SPEC + MANUFACTURING MATRIX

SPEC REF -	MANDREL MND-170905-125.55
ID BAND 125 / 125 / 125	OD BAND 140 / 140 / 140
WALL THICKNESS 7.5 mm	REQUIRED CS 310
NOTCH DIRECTION -	WET RULE Dry + 0.910
TUBE DRY / WET 44.32 / 48.7 g	WEIGHT / MM 0.3247 g

PAPER RECIPE TO FOLLOW

-

-

-

SHIFT

-

Meters Produced	Accepted Meters	Reject Meters	Reject Reason Code

Bamboo equivalent: produced 0 · accepted 0 · reject 0

DIMENSION READINGS

Length	ID	OD	Weight	CS

LINKED REEL ISSUES (OPTIONAL)

4af64cf8 | Shift GENERAL | 50kg

c9787ff6 | Shift GENERAL | 50kg

d8f47067 | Shift GENERAL | 50kg

2d93ba49 | Shift GENERAL | 50kg

9d1ff1e0 | Shift GENERAL | 50kg

00000000 | Shift A | 250kg

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Recipe rows were not captured on this snapshot.

BAMBOO TO BE MADE

TARGET BAMBOO

PCS / BAMBOO

416

BAMBOO LENGTH

USABLE / TRIM

1540 mm1500 / 40 mm

TUBE DRY / WET

BAMBOO DRY / WET

44.32 / 48.7 g709.07 / 779.19 g

GLUE + BRIDGE

Base 15% · Parchment 1.5% · Drying 9%

Paper 115.75 g · Glue 337.5 g · Parchment 33.75 g · Delta -198.55 g

TL-4 (20100) · ratio 20%

Vinsol (30100) · ratio 80%

PARCHEMENT PATTERN

-

- family

NOTCH

Supervisor job card and physical tool snapshot (5 of 5)

Planner assignment is missing for this stage. Assign machine and shift on the planning board before supervisor entry.

Save Draft

Complete Stage

No notch

Distance - - Depth - - Direction -
Blade - - Holder - - Punch -

PACKING

Packed by route stage when required
- - - - / box

BAMBOO MATH

4 bamboo target · 16 pcs/bamboo
1540 mm selected · 1500 mm usable · 40 mm trim

OUTPUT TRUTH

64 open pcs · - produced · - packed

PREVIOUS STAGE ACTUALS

No completed upstream stage is recorded yet.

Release verification

The implementation was checked locally with service, unit, build, static, and focused browser verification. The application is ready for client data onboarding; the final plant acceptance steps are listed below.

Deployment status

The tooling changes are committed and pushed to GitHub. The linked Railway service remains online on its previous deployment. Railway refused the new upload because the account trial has expired; after selecting a Railway plan, redeploy the latest pushed commit and run the authenticated production smoke test.

Verification	Result
Service compilation	Passed
Master tooling contract	5 passed
BFF route/inward/job-card contract	5 passed
Physical lifecycle tests	3 focused tests passed
Web unit/static checks	Passed
TypeScript and production build	Passed
Dependency audit	0 vulnerabilities
Runtime shutdown override	Passed
Focused browser tooling/spec/job-card suite	3 passed
BFF health and service startup	Healthy / ready

Client onboarding acceptance

- Confirm Location Master positions.
- Load approved tool definitions and option values.
- Inward opening physical tool stock.
- Print and apply QR labels.
- Run one supervised issue/return cycle.
- Run one supervised blade grinding cycle.
- Complete one supervised job card and verify the tooling report.

Prepared for client operating use